

CORVALLIS, OR, USA

WMO: 726945

Lat: 44.500N

Lon: 123.283W

Elev: 250

StdP: 14.56

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24202

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.8	27.5	17.5	13.4	28.4	21.4	16.2	31.0	24.6	48.7	22.3	47.8	2.1	300	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.9	92.9	67.2	89.8	66.3	85.5	64.7	69.0	88.8	67.4	86.9	65.7	83.3	9.4	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.3	81.7	75.5	59.4	76.2	74.5	57.4	70.8	71.9	33.1	88.5	31.7	86.5	30.5	83.2	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.8	17.8	16.1	DB	18.2	100.9	6.4	3.4	13.6	103.3	9.8	105.3	6.2	107.2	1.6	109.7
			WB	17.2	72.6	6.2	2.4	12.7	74.4	9.1	75.8	5.6	77.1	1.1	78.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	41.9	43.9	47.7	51.8	58.3	63.1	69.5	69.0	63.7	54.6	46.4	41.4	(6)
	DBStd	11.37	6.35	5.57	5.13	5.21	5.83	5.58	4.97	4.61	5.21	5.48	6.52	6.45	(7)
	HDD50	1027	259	180	105	38	5	0	0	0	0	20	145	275	(8)
	HDD65	4290	716	590	535	397	222	99	14	15	85	326	560	731	(9)
	CDD50	2609	8	10	35	92	261	394	605	590	410	161	35	8	(10)
	CDD65	397	0	0	0	1	13	43	153	140	45	2	0	0	(11)
	CDH74	5746	0	0	1	49	309	717	1953	1811	837	69	0	0	(12)
	CDH80	2343	0	0	0	8	92	276	867	781	307	12	0	0	(13)
(14) Wind	WSAvg	6.2	5.1	5.5	6.2	6.0	6.2	6.7	8.3	7.6	6.3	5.3	5.3	5.5	(14)
(15) Precipitation	PrecAvg	53.8	8.9	6.1	6.0	3.8	2.6	1.5	0.3	0.5	1.5	4.5	8.2	10.0	(15)
	PrecMax	90.1	20.2	17.2	11.3	7.8	5.5	3.2	1.0	2.0	7.0	11.8	22.5	22.2	(16)
	PrecMin	35.4	2.2	0.3	1.1	1.5	0.0	0.2	0.0	0.0	0.1	0.6	1.1	3.8	(17)
	PrecStd	12.4	4.1	3.8	2.5	1.8	1.4	0.7	0.3	0.6	1.5	2.4	4.3	4.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.7	62.9	71.8	81.0	88.2	95.0	99.7	98.8	92.5	81.4	65.7	59.2	(19)
		MCWB	55.1	52.4	56.9	61.9	66.1	67.2	70.2	68.0	64.4	61.3	60.8	54.8	(20)
	2%	DB	56.3	58.1	65.3	72.9	81.8	87.9	92.9	91.5	88.1	74.6	61.4	55.3	(21)
		MCWB	52.0	50.7	53.6	57.8	62.5	65.5	67.7	67.0	64.0	58.5	55.5	51.5	(22)
	5%	DB	54.0	55.2	61.3	67.8	76.6	81.7	89.6	88.2	83.4	70.3	58.2	53.8	(23)
		MCWB	50.1	49.5	51.7	55.3	60.3	63.2	67.0	66.0	63.1	57.5	53.3	50.7	(24)
	10%	DB	51.8	53.7	57.1	63.2	72.0	77.0	84.2	83.8	79.0	65.6	55.4	51.5	(25)
		MCWB	48.5	48.6	49.8	52.6	58.2	60.9	65.0	64.5	61.3	56.3	51.3	48.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.0	55.2	58.4	62.9	67.4	69.9	72.3	70.8	68.3	64.4	61.3	55.7	(27)
		MCDB	59.3	59.4	68.2	78.9	85.7	90.3	94.7	92.2	85.2	75.4	64.8	58.7	(28)
	2%	WB	52.6	52.6	55.2	59.0	63.9	66.5	69.2	68.5	65.9	61.3	57.2	52.6	(29)
		MCDB	55.6	56.6	62.5	71.3	79.7	84.6	89.2	88.6	82.8	70.3	61.0	55.2	(30)
	5%	WB	50.3	50.4	52.9	55.9	61.1	64.0	67.4	66.9	64.0	59.1	54.3	50.3	(31)
		MCDB	53.2	54.6	59.7	66.3	74.9	80.5	87.9	85.7	80.2	66.5	57.6	53.3	(32)
	10%	WB	48.1	48.4	50.9	53.3	58.7	61.7	65.5	65.2	62.1	57.2	52.2	47.7	(33)
		MCDB	51.1	52.5	56.7	61.7	70.8	75.6	83.4	82.4	76.4	63.9	55.5	50.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	14.9	17.6	20.3	22.9	24.5	28.9	29.7	28.3	22.1	14.0	10.7	(35)
		MCDBR	13.2	19.0	24.7	31.1	33.6	34.0	36.6	37.4	38.7	31.7	15.9	11.4	(36)
		MCWBR	10.0	12.9	15.5	18.0	17.8	16.1	16.0	16.4	18.3	18.0	11.6	9.4	(37)
	5% WB	MCDBR	12.2	15.1	22.0	27.9	30.3	31.2	34.2	33.6	32.9	23.8	12.7	10.7	(38)
		MCWBR	9.8	11.1	14.6	16.7	16.7	15.5	15.8	15.5	16.5	14.6	9.9	9.2	(39)
(40) Clear-Sky Solar Irradiance	taub		0.317	0.317	0.324	0.357	0.364	0.358	0.351	0.356	0.345	0.327	0.324	0.327	(40)
	taud		2.534	2.542	2.529	2.398	2.396	2.419	2.457	2.464	2.486	2.548	2.522	2.468	(41)
	Ebn at Noon		257	279	291	287	287	288	289	284	279	271	251	238	(42)
	Edh at Noon		21	25	29	36	37	36	34	33	30	24	21	21	(43)
(44) All-Sky Solar Radiation	RadAvg		415	670	983	1414	1798	1994	2275	1946	1463	864	470	337	(44)
	RadStd		62	85	118	117	172	177	136	122	121	98	61	49	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+31

Nomenclature: See separate page